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Artificial Intelligence for Interpreting Occupational Health Examination Tests: A Systematized Review of Evidence and Gaps

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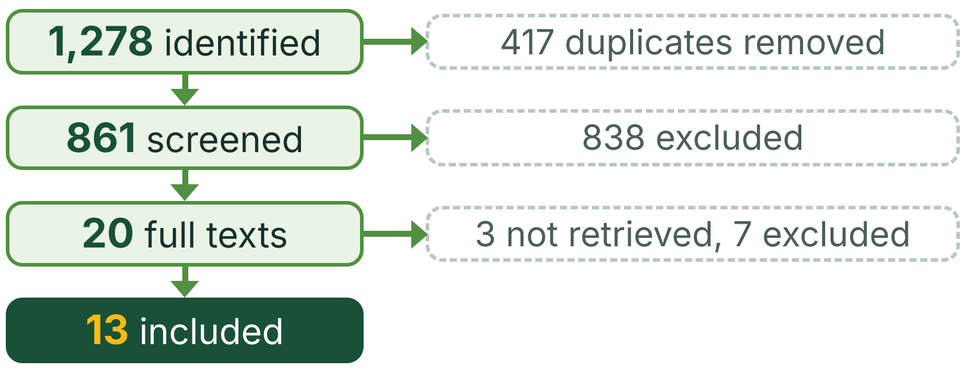
01 Background

- Periodic occupational examinations generate **large volumes of test results**.
- Whether **artificial intelligence (AI)** validly interprets them remains unclear.

Aim: to review the evidence on AI interpreting workers' examination tests against a clinical reference standard.

02 Methods

Design	Systematized review, reported with PRISMA 2020
Databases	PubMed, Scopus, ProQuest, IEEE Xplore
Search	Studies from 2006 onwards
Inclusion	AI interpreting workers' examination tests against a clinical reference standard
Exclusion	Prediction models
Appraisal	JBI diagnostic test accuracy checklist; no pooling



03 Results — Admired behind glass: 0 of 13 evaluated prospectively.

Chest radiography — 9
Reference: expert reading or clinical diagnosis
AUC 0.84–1.00 · Se 62–100% · Sp 46–98%
External validation 3 · Prospective 0

Audiometry — 1
Reference: claimants vs unexposed controls
Se 98.6%, Sp 90.2%
External validation 0 · Prospective 0

Electrocardiography — 1
Reference: examination ECG diagnosis
AUC 0.92–0.98
External validation 0 · Prospective 0

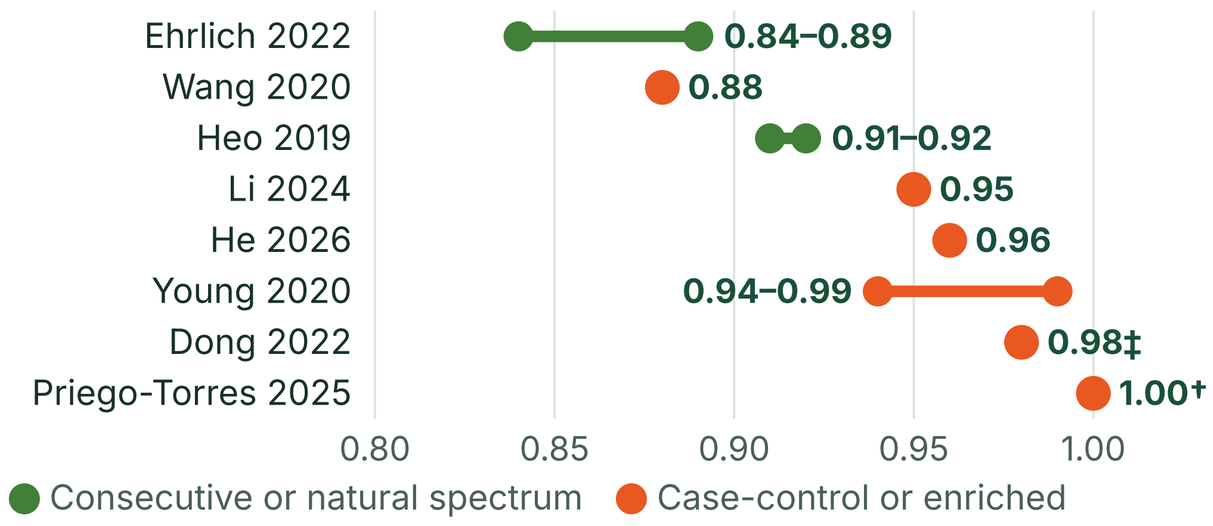


Fitness to work — 0
No study provided AI with the job-demand information required for fitness-to-work decisions

Multimodal records — 2
Reference: physicians' fit/unfit status
AUC 0.60; kappa 1.00 vs one physician rating documents
agreement ≠ accuracy

Spirometry, laboratory — 0
No eligible study: two empty pedestals

Imaging AUC by study design (not pooled)



High AUC ≠ high sensitivity: AUC 0.88 with sensitivity 62%[1]

Case-control and enriched datasets likely overstate screening performance

† Cross-validation, public controls. ‡ Region-level features.

Quality and validation

JBI DIAGNOSTIC TEST ACCURACY CHECKLIST	YES
Q1 Consecutive/random	2
Q2 Case-control avoided	5
Q3 Exclusions appropriate	1
Q4 Index test blinded	11
Q5 Threshold pre-specified	1
Q6 Appropriate reference	6
Q7 Reference blinded	11
Q8 Appropriate interval	12
Q9 Same reference for all	10
Q10 All analysed	7

Yes · Unclear · No · Not applicable

DATA EXTRACTION	YES
External validation	3
NPV reported	2
Prospective evaluation	0

⚠ Limitations of the current evidence

- Evidence centres on chest radiography (9 of 13 studies).
- Clinical-setting spirometry and ECG AI performs well but is unvalidated in workers.[2,3]
- Reference standards are imperfect: human readers agreed only moderately (kappa 0.42–0.66).[1,4]
- One system was deployed in more than 10 hospitals without reported implementation outcomes.[5]

04 Conclusion

Evidence is insufficient to support AI interpretation in periodic examinations. AI is best positioned as decision support under physician responsibility.

Adoption requires:

External validation · **Prevalence-specific NPV** · **Comparison with occupational physicians** · **Prospective evaluation**

Scan: PRISMA flow, 13 included studies, JBI appraisal

References

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